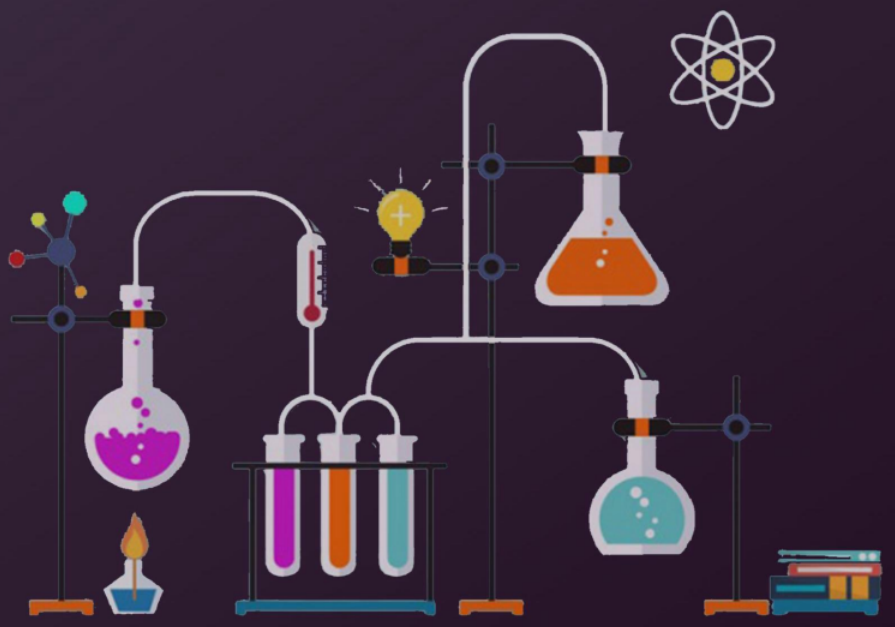
CHEMISTRY

Alcohol Ether & Phenol



SHEET

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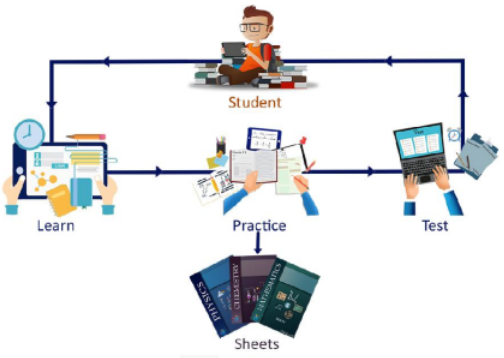
Contents

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2.	Homework Index	Page - iii
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topics	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction to Alcohol, Ether and phenol				
2	Preparation Of Alcohol By Alkenes	1-3		1-3	1-2
3	Preparation From Alkyl Halide, Ether, Esters And Primary Amine	4			8-9
4	Grignard Reagent	5		3-7	3
5	Chemical Properties Of Alcohols	6-12	1-4, 8-9	8-14	14-16
6	Preparation Of Ethers And Their Physical Properties	13-16	10	15-16	
7	Chemical Properties of Ethers	17-20	7	17-20	
8	Epoxy Compounds	21		21	
9	Preparation Methods of Phenol and It's Physical Properties				
10	Reactions of OH Group in Phenols	22		22	
11	Reaction of Benzene Ring in Phenolic Compounds	23-27		23-26	
12	ACC		5-6	27-30	5-7, 10-13, 17

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Preparation Of Alcohol By Alkenes

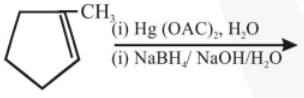
Q1. Bayer's reagent is :-

(A) Alkaline $KMnO_4$ soln.

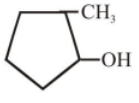
(B) Acidic $KMnO_4$ soln.

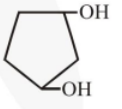
(C) Netural $KMnO_4$ soln.

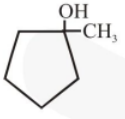
(D) Aqueous bromine soln.

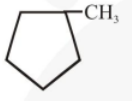
Q2. 

A,A is -

(A) 

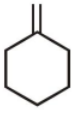
(B) 

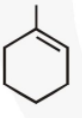
(C) 

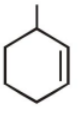
(D) 

Q3. $A \xrightarrow[(ii)\ NaBH_4,\ \bar{O}H]{(i)\ Hg(OAc)_2,\ HOH}$

1–Methylcyclohexanol. Here A is :

(A) 

(B) 

(C) 

(D) (A) or (B)z

Preparation From Alkyl Halide, Ether, Esters And Primary Amine

Q4. Alkyl chloride is formed when alcohol is treated with HCl in presence of anhydrous $ZnCl_2$. The order of reactivity with respect to alcohol is :-

(A) 3° > 2° > 1° (B) 1° > 2° > 3°

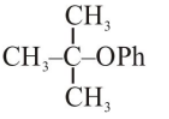
(C) 2° > 1° > 3° (D) 1° > 3° > 2°

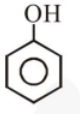
Grignard Reagent

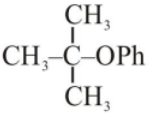
Q5. $Ph–MgBr + CH_3–\underset{\displaystyle CH_3}{\overset{\displaystyle |}{CH}}–OH \longrightarrow A$

What is the A product :-

(A) 

(B) 

(C) 

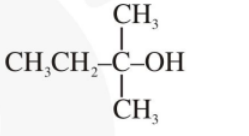
(D) 

Chemical Properties Of Alcohols

Q6. The order of reactivity of following alcohols with halogen acids is _____

(A) $CH_3CH_2–CH_2–OH$

(B) $CH_3CH_2–\underset{\displaystyle CH_3}{\overset{\displaystyle |}{CH}}–OH$

(C) 

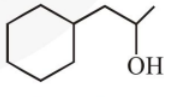
(A) (A) > (B) > (C)

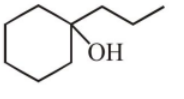
(B) (C) > (B) > (A)

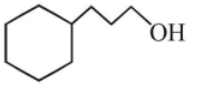
(C) (B) > (A) > (C)

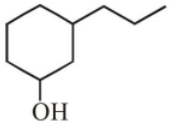
(D) (A) > (C) > (B)

Q7. The alcohol which show turbidity immediate when reacted with ($HCl + ZnCl_2$)

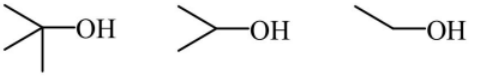
(A) 

(B) 

(C) 

(D) 

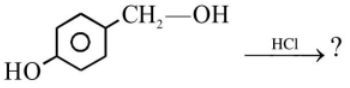
Q8. Identify correct reactivity order for the following compounds on reaction with anhydrous PCl_3



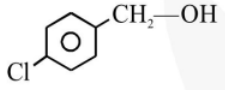
(I) (II) (III)

(A) I > II > III
(B) II > I > III
(C) III > II > I
(D) III > I > II

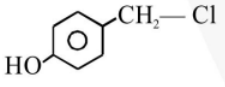
Q9.



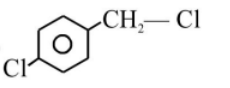
(A)



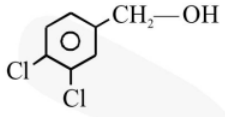
(B)



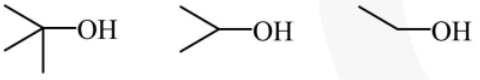
(C)



(D)



Q10. Identify correct reactivity order for the following compounds on reaction with Conc. HCl and hydrous $ZnCl_2$.



(I) (II) (III)

(A) I > II > III
(B) II > I > III
(C) III > II > I
(D) III > I > II

Q11. Lucas reagent is :-

(A) Conc. HCl and anhydrous $ZnCl_2$
(B) Conc. HCl and hydrous $ZnCl_2$
(C) Conc. HNO_3 and hydrous ZnCl
(D) Conc. HNO_3 and anhydrous $ZnCl_2$

Q12. Alcohol which gives blue colouration in victor meyer test is :-

(A) Only CH_3CH_2—OH
(B) Only CH_3CH_2—CH_2OH
(C) All alcohols containing —CH_2—OH group
(D) All alcohols containing >CH—OH group

Preparation Of Ethers And Their Physical Properties

Q13. Williamson's synthesis involves :-

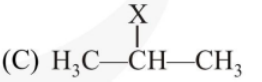
(A) S_N1 mechanism
(B) Nucleophilic addition
(C) S_N2 mechanism
(D) S_E mechanism

Q14. In the williamson synthesis of ethers given by the general equation
R–X + R'–ONa ⟶ R–OR', the yield form R–X follows the sequence :-

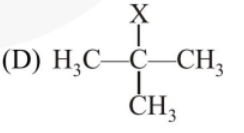
(A) CH_3 > 1° > 2° > 3°
(B) CH_3 < 1° < 2° < 3°
(C) CH_3 < 1° > 2° > 3°
(D) CH_3 > 1° < 2° > 3°

Q15. Sodium tertiary butoxide forms ether only with :

(A) CH_3—CH_2—Br
(B) CH_3—X



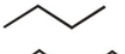
(C)

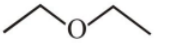


(D)

Q16. Find out correct product of following reaction.

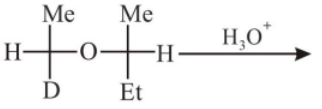
$$CH_3CH_2Br \xrightarrow{Ag_2O,\ dry}$$

(A)  (B) $CH_2 = CH_2$

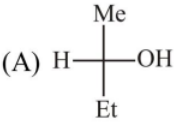
(C)  (D) CH_3CH_2OH

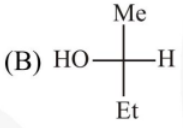
Chemical Properties of Ethers

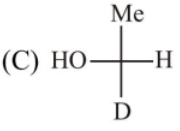
Q17. Which of the following alcohol is not present in the mixture containing major products.

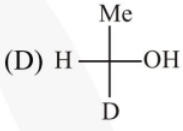


Product mixture.

(A) 

(B) 

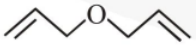
(C) 

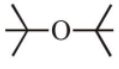
(D) 

Q18. Which of the following ethers is the most unreactive to cleavage with conc. HBr ?

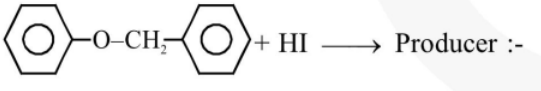
(A) Ph—CH_2—O—CH_3

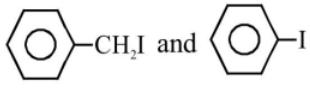
(B) Ph—O—Ph

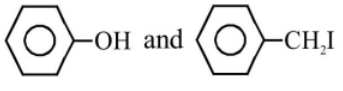
(C) 

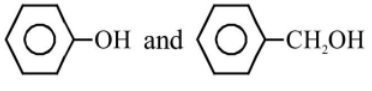
(D) 

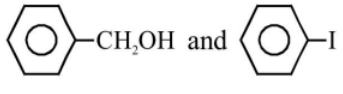
Q19. The reaction :



(A) 

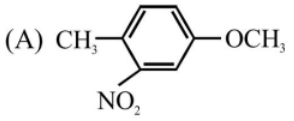
(B) 

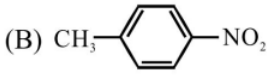
(C) 

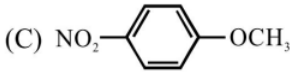
(D) 

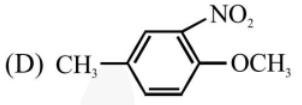
Q20. 

Major product is :-

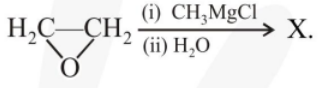
(A) 

(B) 

(C) 

(D) 

Epoxy Compounds

Q21. 

The product obtained in this reaction is :

(A) CH_3CH_2OH

(B) $(CH_3)_2CHOH$

(C) $CH_3CH_2CH_2OH$

(D) HO—CH_2—CH_2—CH_2—CH_2—OH

Reactions of OH Group in Phenols

Q22. Benzoylation of phenol with benzoyl chloride in the presence of dilute NaOH gives phenyl benzoate. This reaction is an example of

(A) Friedel – Crafts reaction

(B) Reimer – Tiemann reaction

(C) Claisen – Schmidt reaction

(D) Schotten – Baumann reaction

Reaction of Benzene Ring in Phenolic Compounds

Q23. Phenol reacts with bromine in CS_2 at low temperature to give :-

(A) m–bromophenol

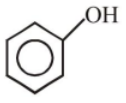
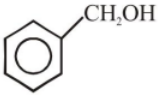
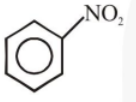
(B) p–bromophenol

(C) o & p–bromophenol

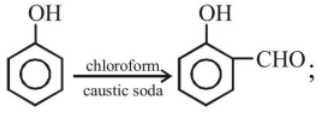
(D) 2,4,6-Tribromophenol

Q24. Phenol on sulphonation gives :-
(A) o–phenol sulphonic acid
(B) p–phenol sulphonic acid
(C) m–phenol sulphonic acid
(D) o–and p–phenol sulphonic acid

Q25. Which one of the following is most reactive towards electrophilic attack :-

(A) 
(B) 
(C) 
(D) 

Q26. Phenol gives characteristic colouration with :-
(A) Iodine solution
(B) Bromine water
(C) Aqueous $FeCl_3$ solution
(D) NH_4OH

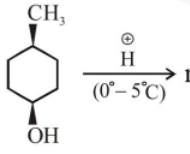
Q27. 

The above reaction is :-
(A) Electrophilic substitution
(B) Nucleophilic substitution
(C) Free radical addition
(D) Electrophilic addition

Exercise - 1A *Concept Builder*

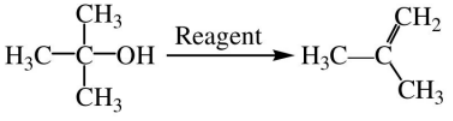
ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1.  major products obtained are

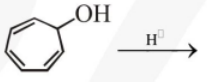
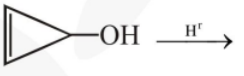
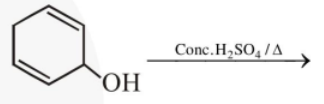
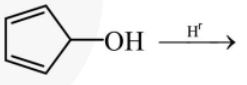
(A) Racemic mixture
(B) Diastereomers
(C) Geometrical isomers
(D) Positonal isomers

Q2. Identify reagents for following reaction sequence -

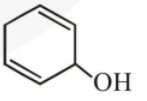
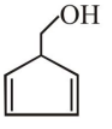
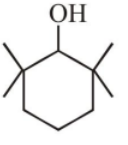


(A) conc.HI
(B) conc. H_2SO_4/Δ
(C) conc. H_3PO_4/Δ
(D) conc. HBr

Q3. Which of the reaction(s) among following produces aromatic compound ?

(A) 
(B) 
(C) 
(D) 

Q4. Which alcohols on reaction with concentrated H_2SO_4 at high temperature will give product, which give +ve test with Br_2 solution :

(A) 
(B) 
(C) $Me_3C–CH_2–CH_2OH$
(D) 

MATCHING LIST TYPE

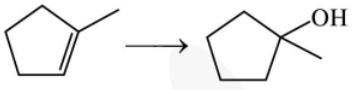
*Each question has TWO (02) **matching lists: LIST I and LIST II**. OUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching.For each question, choose the option corresponding to the correct matching.*

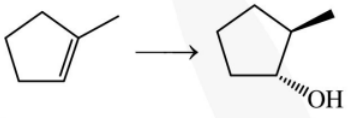
Q5. Match the chemical conversions in List-I with the appropriate reagents in List-II and select the correct answer using the code given below this lists -

List-I **List-II**

(P) 

(Q) $H_3C-C\equiv CH \longrightarrow H_3C-\underset{\underset{O}{\|}}{C}-CH_3$

(R) 

(S) 

(1) (i) $Hg(OAc)_2$, H_2O (ii) $NaBH_4$

(2) Conc. H_2SO_4 , Δ

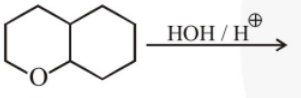
(3) $HgSO_4$ + 1% H_2SO_4

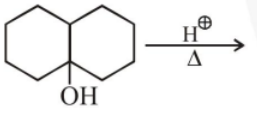
(4) (i) B_2H_6(THF) (ii) H_2O_2 /NaOH

Codes :

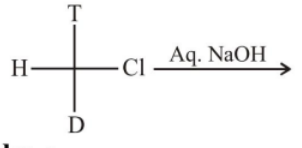
	P	Q	R	S
(A)	2	3	1	4
(B)	3	2	1	4
(C)	2	3	4	1
(D)	3	2	4	1

Q6. **List-I (Reaction)** **List-II (Mechanism)**

(P) 

(Q) 

(R) $CH_3-CH_2-\overset{\ominus}{O}Na^{\oplus} + H_3C-\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}-Cl \longrightarrow$

(S) 

(1) S_N2

(2) E^2

(3) S_N1

(4) E^1

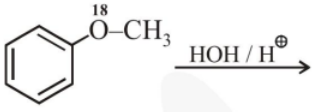
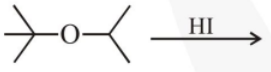
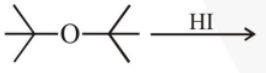
Codes :

	P	Q	R	S
(A)	1	2	3	4
(B)	4	3	2	1
(C)	3	2	4	1
(D)	3	4	2	1

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

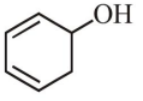
Q7.

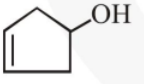
Column-I (Reaction)	**Column-II (Molecular weight of aliphatic alcohol formed as major product)**
(A) $CH_3–CH_2–O–CH_3 \xrightarrow{\text{Anhydrous HI}}$	(P) 32
(B) 	(Q) 46
(C) 	(R) 74
(D) 	(S) 60

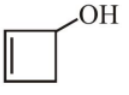
NUMERICAL TYPE

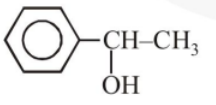
*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

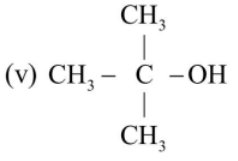
Q8. Which of following shows fast rate of dehydration in compare with 2-butanol :

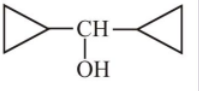
(i) 

(ii) 

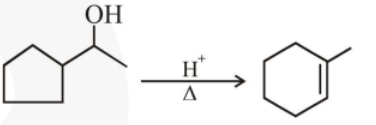
(iii) 

(iv) 

(v) 

(vi) 

Q9. In the following reaction



Find out number of 1,2-shift of the carbocation formed :

Q10. $Me_2CHCH_2OH \xrightarrow{Na} A \xrightarrow{MeI} B$

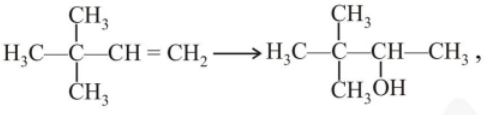
If molecular weight of A (sodium alkoxide) is X_1 & molecular weight of B is X_2.

Then the value of $(X_1 – X_2)$ is :

Exercise - 2 — Brain Booster

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. 

This interconversion can be achieved by :

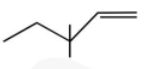
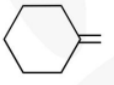
(A) Acid catalysed hydration

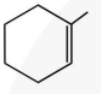
(B) Oxymercuration-demercuration

(C) Hydroboration - oxidation

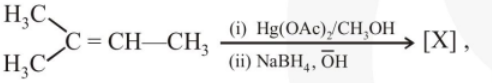
(D) Any method mentioned above

Q2. HBO, oxymercuration-demercuration and acid catalysed hydration will give same product in

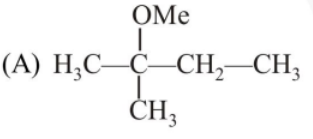
(A)  (B) 

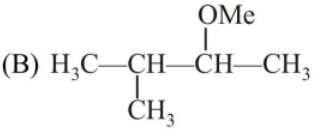
(C)  (D) 

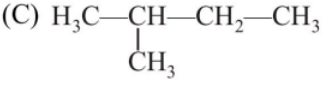
Q3. In the given reaction

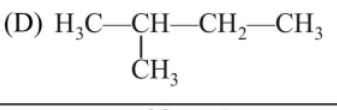


[X] will be :

(A) 

(B) 

(C) 

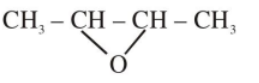
(D) 

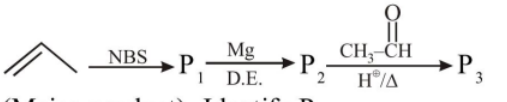
Q4. Which of the following compound gives tertiary alcohol with CH_3MgBr :

(A) C_6H_5CHO

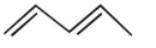
(B) $C_2H_5CO_2CH_3$

(C) C_2H_5COOH

(D) 

Q5. 

(Major product). Identify P_3

(A)  (B) 

(C)  (D) 

Q6. 2-Phenylbutan-2-ol can be prepared by which of the following combinations followed by :

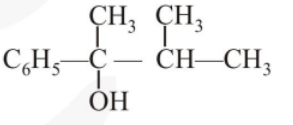
(A) $C_6H_5COCH_3 + C_2H_5MgBr$

(B) $C_2H_5COCH_3 + C_6H_5MgBr$

(C) $C_6H_5COC_2H_5 + CH_3MgBr$

(D) All of these

Q7. Which set of the following reagents (A to D) would you select to convert $C_6H_5COCH_3$ (acetophenone) to the following alcohol ?



(A) $CH_3CH_2CH_2MgBr$ and hydrolysis

(B) $CH_3CH(Br)CH_3.AlCl_3$

(C) $(CH_3)_2CHMgBr$ and acid hydrolysis

(D) $CH_3CHCHCH_3$, Zn

Q8. The order of ease of dehydration of alcohol is:-

(A) p-Alcohol > s-alcohol > t-alcohol

(B) p-Alcohol > t-alcohol > s-alcohol

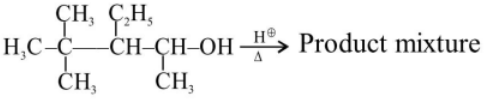
(C) t-Alcohol > s-alcohol > p-alcohol

(D) t-Alcohol > p-alcohol > s-alcohol

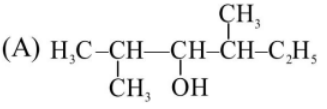
Q9. To distinguish between primary, secondary and tertiary alcohol, the test adopted is :-

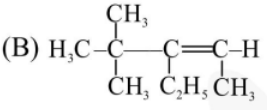
(A) Oxidation test (B) Lucas test

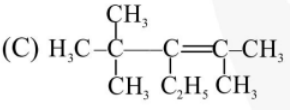
(C) Victor-Meyer test (D) All the above

Q10.  Product mixture

the products mixture contains :

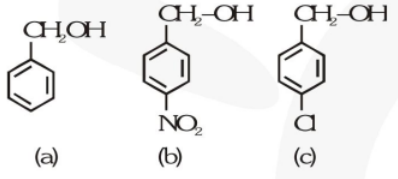
(A) 

(B) 

(C) 

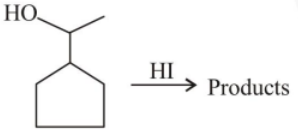
(D) None of these

Q11. Mark the correct increasing order of reactivity of the following compounds with HBr/HCl :

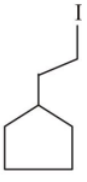
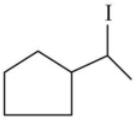


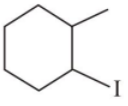
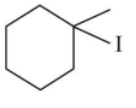
(A) a < b < c (B) b < a < c

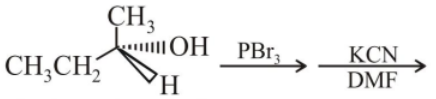
(C) b < c < a (D) c < b < a

Q12. 

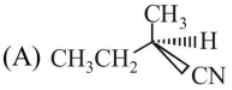
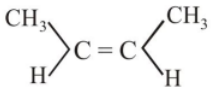
Which of the following will be not fitted as product.

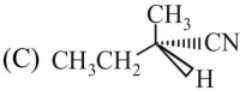
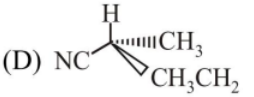
(A)  (B) 

(C)  (D) 

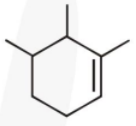
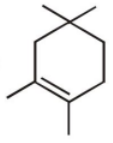
Q13. 

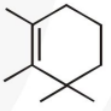
Final product is ?

(A)  (B) 

(C)  (D) 

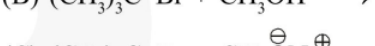
Q14. Treatment of 2,2,6,6-tetramethyl-cyclohexanol with acids at high temperature leads to formation of alkene after rearrangment. The strcuture of major alkene is :

(A)  (B) 

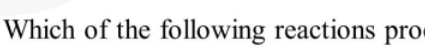
(C)  (D) 

Q15. Which of the following reactions would give the best yield of t-butyl methyl ether ?

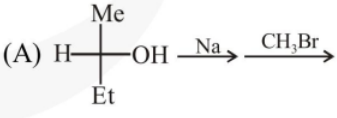
(A) 

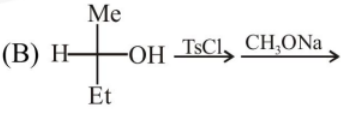
(B) 

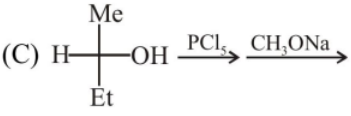
(C) 

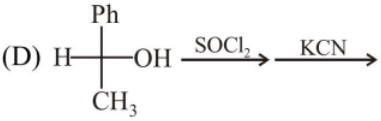
(D) 

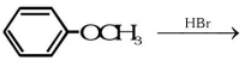
Q16. Which of the following reactions proceeds with retention of configuration ?

(A) 

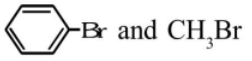
(B) 

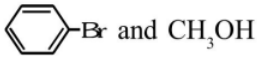
(C) 

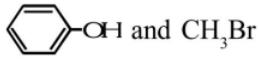
(D) 

Q17. In the reaction  the products are :-

(A)  and H_2

(B)  and CH_3Br

(C)  and CH_3OH

(D)  and CH_3Br

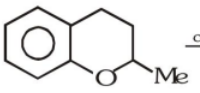
Q18. A on treatment with Na gives B and with PCl_5 gives C. B and C react together to give diethyl ether. A, B and C are in the order :-

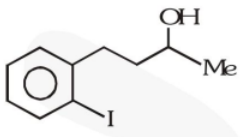
(A) C_2H_5OH, C_2H_5Cl, C_2H_5ONa

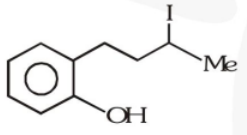
(B) C_2H_5OH, C_2H_6, C_2H_5Cl

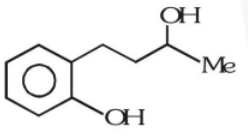
(C) C_2H_5OH, C_2H_5ONa, C_2H_5Cl

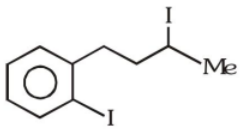
(D) C_2H_5Cl, C_2H_6, C_2H_5OH

Q19.  Product ; Product is:

(A) 

(B) 

(C) 

(D) 

Q20. In following reaction product is :

$Me_3C—O—CH_2CH_3 + \underset{(1\ Mole)}{HI} \xrightarrow{\Delta}$

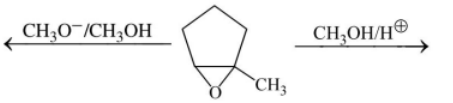
Product

(A) $Me_3C—OH + CH_3CH_2I$

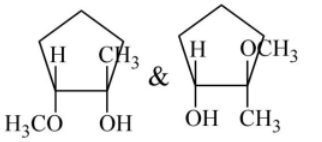
(B) $Me_3C—I + CH_3CH_2OH$

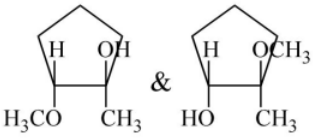
(C) $Me_3C—I + CH_3CH_2I$

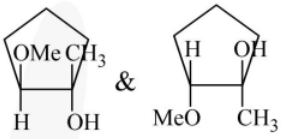
(D) $Me_3C—OH + CH_3CH_2OH$

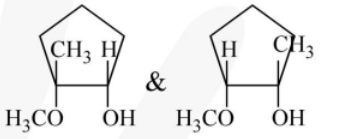
Q21. (A) 

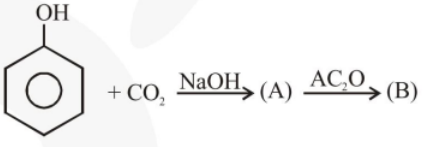
(B) A & B are respectively ?

(A) 

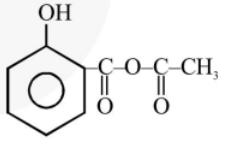
(B) 

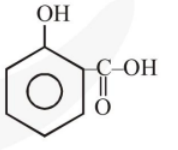
(C) 

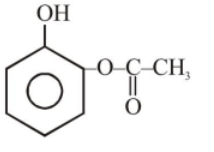
(D) 

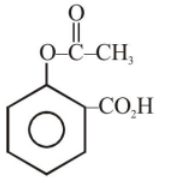
Q22. 

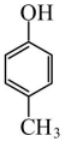
Product (B) will

(A) 

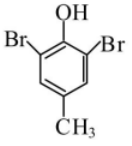
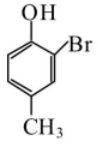
(B) 

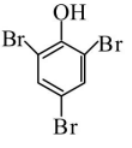
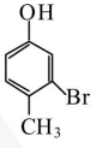
(C) 

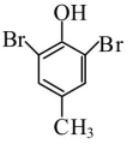
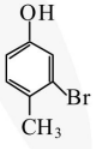
(D) 

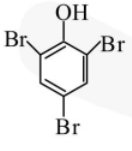
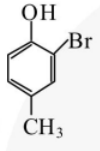
Q23. A $\xleftarrow{Br_2, H_2O}$  $\xrightarrow{Br_2, CS_2}$ B (major)

(major) 'A' and 'B' are respectively :

(A)  

(B)  

(C)  

(D)  

Q24. Which of the following reacts with phenol to give salicylaldehyde after hydrolysis :-

(A) Dichloro methane

(B) Trichloro methane

(C) Methyl chloride

(D) None of these

Q25. Phenol $\xrightarrow[2.\ CO_2/140°C]{1.\ NaOH}$ A $\xrightarrow{H^+/H_2O}$ B $\xrightarrow{Ac_2O}$ C

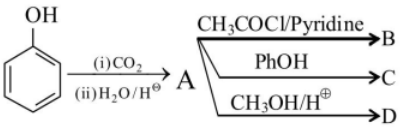
In this reaction, the end product C is :

(A) salicylaldehyde

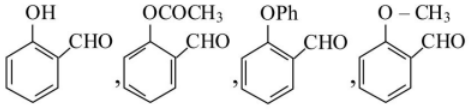
(B) salicylic acid

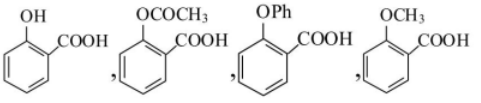
(C) phenyl acetate

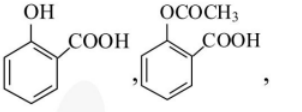
(D) aspirin

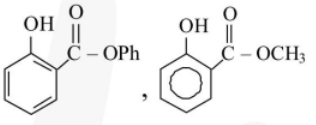
Q26. 

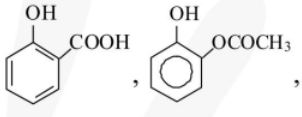
A, B, C & D are respectively :-

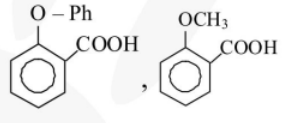
(A) 

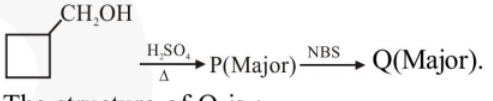
(B) 

(C) 

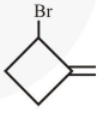
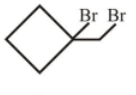


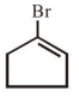
(D) 



Q27. 

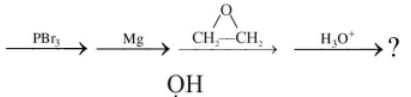
The structure of Q is :-

(A)  (B) 

(C)  (D) 

Q28. What is the major organic product of the following sequence of reaction ?

$(CH_3)_2CHCH_2–OH$



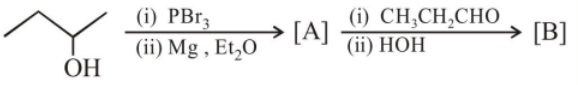
(A) $(CH_3)_2CH–\underset{OH}{\underset{|}{C}}HCH_2CH_3$

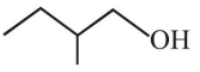
(B) $(CH_3)_2CHCH_2\underset{OH}{\underset{|}{C}}H–CH_3$

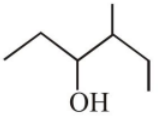
(C) $(CH_3)_2CHCH_2CH_2–OH$

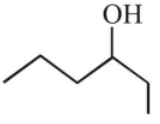
(D) $(CH_3)_2CHCH_2CH_2CH_2OH$

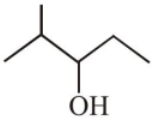
Q29. The correct structure for compound B will be :

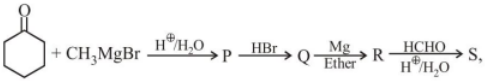


(A) 

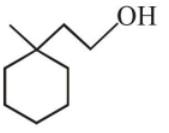
(B) 

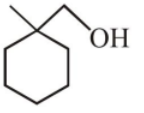
(C) 

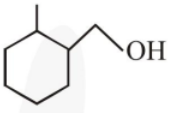
(D) 

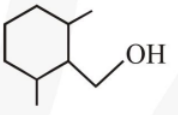
Q30. 

S is :

(A) 

(B) 

(C) 

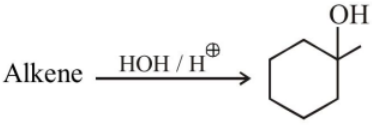
(D) 

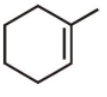
Exercise - 2A — *Brain Booster*

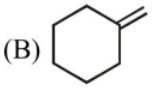
ONE OR MORE THAN ONE CORRECT TYPE

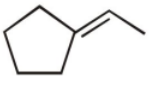
Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

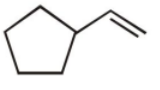
Q1. Which of the following alkene (C_7H_{12}) gives 1-methyl cyclo hexanol as major product :



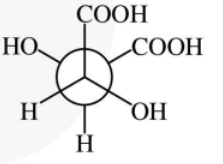
(A) 

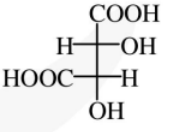
(B) 

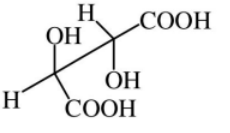
(C) 

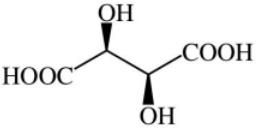
(D) 

Q2. Fumeric acid $\xrightarrow[\text{cold.KMnO}_4]{\text{1\% Alkaline}}$ Product

(A) 

(B) 

(C) 

(D) 

Q3. End-product of which of following reaction will give 2° alcohol as one of the product

(A) $H-\overset{\overset{O}{\|}}{C}-Cl \xrightarrow[(ii)\,H^{\oplus}]{(i)\,CH_3MgBr(excess)}$

(B) $Ph-\overset{\overset{O}{\|}}{C}-O-Et \xrightarrow[(ii)\,H^{\oplus}]{(i)\,CH_3MgBr(excess)}$

(C) $H-\overset{\overset{O}{\|}}{C}-O-Et \xrightarrow[(ii)\,H^{\oplus}]{(i)\,CH_3MgBr(excess)}$

(D) $CH_3-\overset{\overset{O}{\|}}{C}-H \xrightarrow[(ii)\,H^{\oplus}]{(i)\,CH_3MgBr(excess)}$

Q4. Select ether(s) which can be prepared by direct Williamson's ether synthesis.

(A) $CH_3CH_2CH_2–O–CH_2CH_2CH_3$

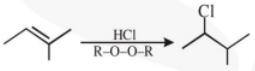
(B) $Ph–O–CH_2CH_3$

(C) $(CH_3)_3C–O–C_2H_5$

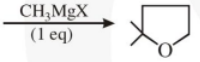
(D) $CH_3CH=CH–O–CH=CH_2$

Q5. Which of the following reactions product is correctly matched.

(A) Isobutane + Cl_2 / hν $\rightarrow$ tertiary butyl chloride (Major product)

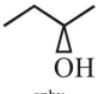
(B)  (Major product)

(C) $CH_3-\overset{\overset{O}{\|}}{C}-CH_2CH_2CH_2-Cl$



(D) R–2–butanol $\xrightarrow[ii)\ CH_3I]{i)\ Na}$ R-2-methoxybutane

Q6. Correct statement for the given sequence of reaction is/are :

 $\xrightarrow{NaH}$ (P) $\xrightarrow{CH_3I}$ (Q) $\xrightarrow[HI]{anhy}$ (A) + (B)

Alcohol

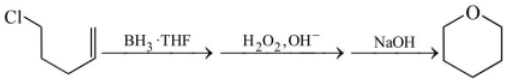
(A) The conversion from (P) to (Q) is S_N2 reaction

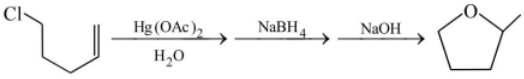
(B) The conversion from (Q) to (A + B) is S_N2 reaction

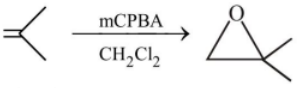
(C) The configuration of product Q is 'R'

(D) (A) on reaction with Na/dry ether give C_2H_6

Q7. Select the correct synthesis

(A) 

(B) 

(C) 

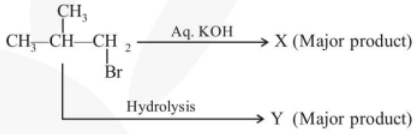
(D) 

PARAGRAPH TYPE

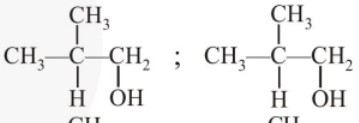
*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) corresponds to the correct answer. For each question, choose the option(s) corresponding to the correct answer.*

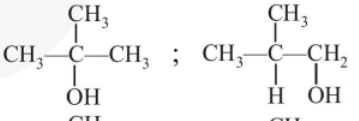
Paragraph for Q. 08 to 09

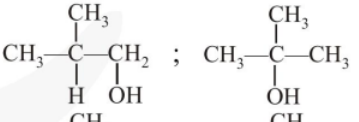
Alkyl halides gives alcohol by SN^1, SN^2 depending on various conditions.

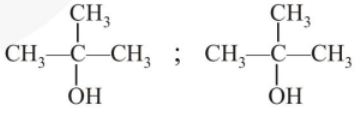


Q8. X & Y are respectively :

(A) 

(B) 

(C) 

(D) 

Q9. Correct statement(s) above reactions is/are :

(A) Major product X is formed by S_N2 mechanism

(B) Major product Y is formed by S_N1 mechanism

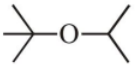
(C) Major product X is formed by S_N1 mechanism

(D) Major product Y is formed by S_N2 mechanism

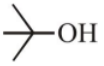
MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

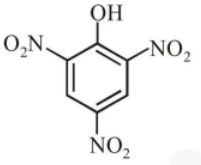
Q10. **List-I (Compound)** **List-II**

(P) 

(1) It gives immidiate turbidity with lucas reagent

(Q) 

(2) It gives yellow ppt. with NaOH / I_2

(R) 

(3) It release CO_2 with $NaHCO_3$

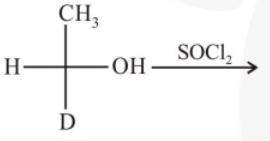
(S) 

(4) It gives 2° alcohol as one of the product with conc. HI

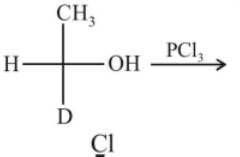
Codes :

	P	Q	R	S
(A)	4	2	3	1
(B)	4	1	3	2
(C)	1	2	3	4
(D)	1	4	2	3

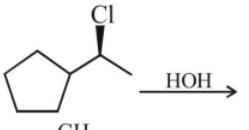
Q11. **List-I (Reaction)** **List-II (For major product)**

(P) 

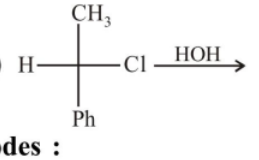
(1) Inversion of configuration takes place

(Q) 

(2) Retension of configuration takes place

(R) 

(3) Racemisation takes place

(S) 

(4) Product is optically inactive

Codes :

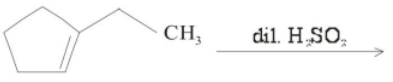
	P	Q	R	S
(A)	2	1	4	3
(B)	1	2	3	4
(C)	2	1	3	4
(D)	2	3	4	1

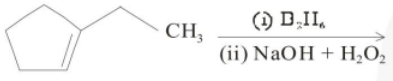
MATCH THE COLUMN TYPE

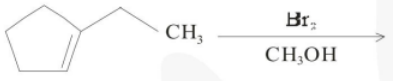
*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

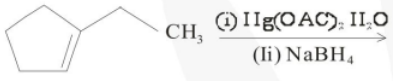
12.

Column-I

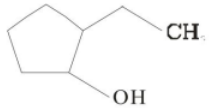
(A) 

(B) 

(C) 

(D) 

Column-II

(P) 

(Q) Product contain atleast one chiral carbon

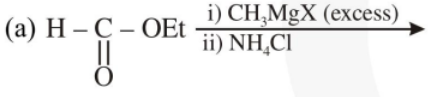
(R) A positive ion participate in mechanism

(S) Net reaction follows markonikow rule

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

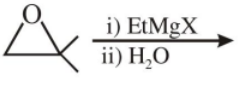
Q13. How many reactions giving 2° alcohol as one of the product.

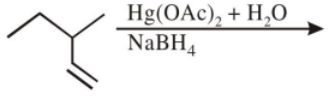
(a) 

(b) Ethyl chloride $\xrightarrow[KOH]{\text{alcoholic}}$

(c) $Me_2CH - O - CH_3 \xrightarrow{HI}$

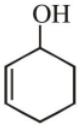
(d) Ethanoic acid $\xrightarrow[ii)\ H_2O]{i)\ LiAlH_4}$

(e) 

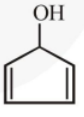
(f) 

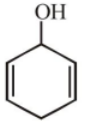
(g) Isopropyl chloride $\xrightarrow[KOH]{\text{aqueous}}$

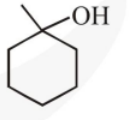
Q14. How many of the following compounds will give immidiate turbidity with Lucas reagent [anhydrous $ZnCl_2$/conc.HCl].

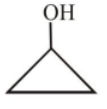
(A) CH_2=CH–OH (B) 

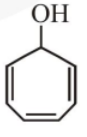
(C) CH_3–OH (D) 

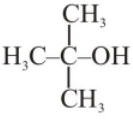
(E) 

(F) 

(G) 

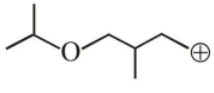
(H) 

(I) 

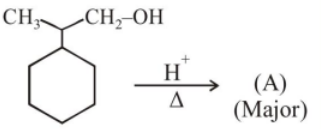
(J) 

(K) H_2C=CH–CH_2–OH

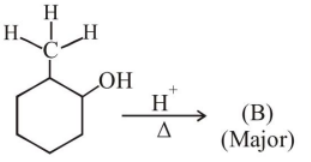
Q15. Number of 1, 2-shift during rearrangement of following carbocation



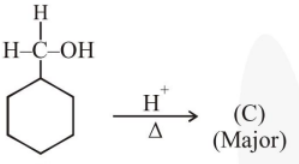
Q16. **Reaction - I :**



Reaction - II :



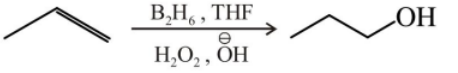
Reaction - III :

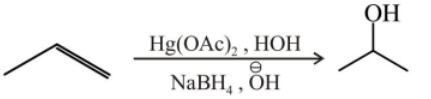


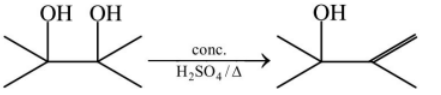
If number of α-Hydrogen in products A, B, C are X_1, X_2 & X_3 respectively :

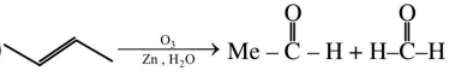
Then the value of $(2X_1 - X_2 - X_3)$ is :

Q17. Among the following reactions identify total number of reactions correctly matched with their major products.

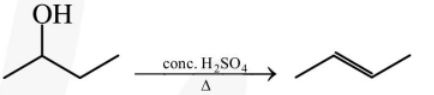
(A) 

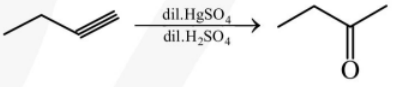
(B) 

(C) 

(D) 

(E) 

(F) 

(G) 

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
A	C	D	A	A	B	B	C	B	A
11	12	13	14	15	16	17	18	19	20
A	D	C	A	B	C	C	B	B	D
21	22	23	24	25	26	27			
C	D	C	D	A	C	A			

Ex-1A

1. B, C **2.** B, C **3.** A, B, C **4.** C, D **5.** A

6. D **7.** (A)→Q ; (B)→P ; (C)→S ; (D)→R **8.** 4

9. 2 **10.** 8

Ex-2									
1	2	3	4	5	6	7	8	9	10
B	C	A	B	B	D	C	C	D	B
11	12	13	14	15	16	17	18	19	20
C	A	C	C	D	A	D	C	B	B
21	22	23	24	25	26	27	28	29	30
B	D	A	B	D	C	C	D	B	B

Ex-2A

1. A, B, D **2.** A, C, D **3.** A, C, D **4.** A, B, C **5.** C, D

6. A, B, C, D **7.** A, B, C **8.** C **9.** A, B **10.** B

11. A **12.** A→R, S, B →P, Q, C→ S, D → S **13.** 4

14. 7 **15.** 3 **16.** 6 **17.** 5